STRATUM

From Lean Canvas to the Innovation Summit, and what comes after

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VERSION	DATE	COVERS	NEXT REVIEW
1.0	12 October 2026	Oct 2026 – mid-2027	22 Nov 2026

1 How to use this plan

The plan is split into seven phases. Each phase has one goal, a short task list with an owner for every task, a clear "done when" test, and a checkpoint with Dr. Pant at the end. A phase is finished when its checkpoint is passed, not when its dates run out.

Owners are written as initials: SV Shrugal, AH Aditya, IK Ishaan, AS Abeer, VP Dr. Pant. The owner does not have to do the whole task alone. The owner is the person who makes sure it gets done and says so in the weekly meeting. Roles are a first proposal; swap them at the first team meeting if your strengths fall differently.

2 The competition at a glance

Dates and requirements below are from the official 2026–2027 Conrad Challenge Student Guide. Re-check the guide before each deadline in case anything changes.

STAGE	DEADLINE	WHAT IS DUE
Activation Stage	29 Oct 2026, 11:59 pm CT	Registration of all members and the coach (over 18) in the Award Force portal; Lean Canvas with 12 answers; final team and innovation name; one category
Innovation Stage (online)	7 Jan 2027, 11:59 pm CT	Innovation Brief (10 sections, 3,000 words total, plus a reference attachment); Innovation Video of 3–5 minutes showing an Innovation Model; Innovation Website; entry fee of US$499 (financial aid available)
Finalists announced	February 2027	Judges' coaching comments released at the same time
Innovation Summit	21–24 Apr 2027	Power Pitch to judges and a public Innovation Expo at Space Center Houston and NASA Johnson Space Center

Time-zone check. 11:59 pm Central Time is 11:29 am the next day in India (10:29 am while US daylight saving is on). Our internal deadlines below are always two days earlier than the official ones.

How the Innovation Stage is judged

CATEGORY	WEIGHT	WHAT STRATUM HAS TO SHOW
Innovation	30%	The correction idea is new, the India gap is real, and we have a plan to verify claims (pre-registered hypotheses, labelled satellite data)

CATEGORY	WEIGHT	WHAT STRATUM HAS TO SHOW
Storytelling and professionalism	20%	One clear story across brief, video and website; the same numbers everywhere; a polished, consistent look
Practicality	20%	A working v0 prototype, a data coverage audit, free data and software, realistic next steps
Marketing strategy	20%	Specific early adopters, a believable route to airlines, awareness of Google, Breakthrough Energy and commercial contrail services
Finances	10%	Honest cost table, simple revenue assumptions, a funding plan that does not depend on luck

3 Timeline

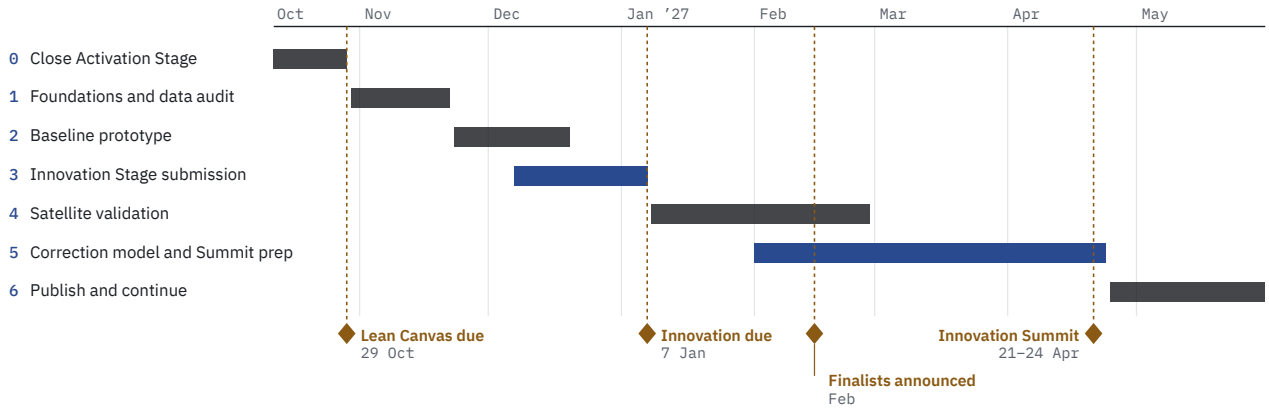


Figure 1. Phases against the official deadlines. Phase 3 starts before Phase 2 ends on purpose: writing the brief early shows us which results we still need. Blue bars are competition submissions.

4 Team roles and working rhythm

MEMBER	ROLE	OWNS
Shrugal Vinod	Modelling lead	Physics baseline, correction model, evaluation, hypotheses, research paper
Aditya Harsoda	Flight data lead	ADS-B data, trajectory pipeline, coverage audit, fuel estimates, code repository
Ishaan Kupnur	Satellite lead	Meteosat downloads, contrail detection, labelling guide and labelled set
Abeer Shah	Product and business lead	Dashboard, website, video, market and competitor research, outreach
Dr. Varsha Pant	Supervisor and coach	Checkpoints, scientific review, introductions, portal registration as coach

Weekly rhythm

- **Sunday, 45 min call.** Each person: what I finished, what I am doing next, what is blocking me. Update the task board during the call.
- **Wednesday, written check-in.** Three lines each in the group chat. No meeting.
- **End of each phase.** 30-minute checkpoint with Dr. Pant.

Shared tools

- **One code repository** with a README that says how to run everything.
- **One shared folder** for documents, figures and the labelled data.
- **A task board** with three columns: to do, doing, done.
- **A decision log.** One line per decision, with the date and the reason. Judges ask "why did you choose X?"

0 Close the Activation Stage

12 – 29 Oct 2026
internal deadline 27 Oct

GOAL	DELIVERABLE	CHECKPOINT
A complete, consistent Activation Stage submission, and a team that is set up to work.	Submitted Lean Canvas with the canvas PDF attached; repository and shared folder live.	Submission confirmation received by 27 Oct.

TASK	OWNER	DONE WHEN
In the Additional Information box, change "roughly two-thirds of aviation's total climate warming" to "more than half", so it matches the Problem answer and the literature	SV	Text updated before submitting
Check every member and every guardian has accepted the Student Guide, Rules and Terms in the portal	AS	All names show as complete
Confirm Dr. Pant is registered as coach under the team account	VP	Coach shows in the Registration tab
Make sure the team name and innovation name match exactly in the Registration and Lean Canvas tabs	AS	Both read "STRATUM"
Upload the Lean Canvas PDF and a team photo in Photos and Files, then submit	SV	Confirmation email saved in the shared folder
Create the code repository, shared folder, task board and decision log	AH	Everyone has access
First team meeting: agree roles, the weekly call time, and this plan	All	Roles table above updated
Everyone reads the Project Synopsis and two papers: Teoh et al. 2024 and Sonabend-W et al. 2024	All	Each person can explain in two minutes why contrails matter and why humidity is the hard part

1 Foundations and data audit

30 Oct – 22 Nov 2026
about 3.5 weeks

GOAL	DELIVERABLE	CHECKPOINT
Every data source opened and tested, the physics reproduced, and a firm answer on whether the flight data covers India well enough.	A two-page coverage audit memo and three working notebooks (ERA5, ADS-B, Meteosat).	22 Nov with Dr. Pant: choose the flight data source using the rule in the research paper (Section 3.3).

Accounts and data access (start in week one, some take time)

TASK	OWNER	DONE WHEN
Create a Copernicus Climate Data Store account, accept the ERA5 licence, download one day of pressure-level data over India (400–150 hPa)	SV	File opens in xarray; temperature at 250 hPa plotted
Request OpenSky historical data access for research, with Dr. Pant's support and a short project description	AH, VP	Request sent; reply tracked in the decision log
Download the ADS-B Exchange free sample day (1 November 2026) and extract flights inside 6–37°N, 68–98°E	AH	Number of flights above FL250 counted
Create a EUMETSAT Data Store account and download Meteosat-9 IODC images for one day	IK	10.8 – 12.0 µm difference image made for one scene over India
Install pycontrails and run its CoCiP example notebook	SV, AH	Example runs end to end on our laptops or Colab

Science and audit

TASK	OWNER	DONE WHEN
Reproduce the Schmidt–Appleman diagram (Figure 1 of the paper) from our own code	SV	Threshold of −41.9 °C at 250 hPa, $\eta = 0.3$
Coverage audit: for sample days, count Delhi–Mumbai flights with continuous cruise tracks and compare with scheduled flights on the route	AH	Percentage captured, per source, written in the memo
Find contrails by eye in 10 Meteosat scenes over India and save screenshots	IK	At least a few clear examples, or a note that they are hard to see and why
Weekly reading group: one paper per week, presented in 10 minutes by one person (see Appendix A)	All	Three papers presented

People and market

TASK	OWNER	DONE WHEN
Draft an advisor email (five sentences plus the synopsis) and send it to 4–6 atmospheric scientists, for example at IISc's Centre for Atmospheric and Oceanic Sciences, IITM Pune, IIT Delhi's Centre for Atmospheric Sciences and IIT Bombay's climate studies programme, plus one contrail group abroad	AS, SV, VP	Emails sent; follow-up date set for one week later
Market sheet: Indian passenger and cargo airlines, fleet size, any published climate targets	AS	Sheet with sources for every number

TASK	OWNER	DONE WHEN
Competitor sheet: Google's contrail work, Breakthrough Energy and pycontrails, commercial contrail services; what each covers and where	AS	One line on why each does not yet solve the problem over India

If the audit says OpenSky coverage is poor: do not lose time trying to fix it. Switch to the twelve free ADS-B Exchange days a year, record the decision in the log, and carry on. Fewer days of good data beat more days of bad data.

2 Baseline prototype (v0)

23 Nov – 20 Dec 2026
4 weeks

GOAL	DELIVERABLE	CHECKPOINT
A physics-only contrail map for the Delhi–Mumbai corridor for January 2026 that anyone can open in a browser.	Prototype v0 link and three overlay images (predicted contrails drawn on satellite images).	20 Dec: live demo to Dr. Pant. This is the Innovation Model for the video.

TASK	OWNER	DONE WHEN
Trajectory pipeline: filter to cruise, remove bad points, resample to one-minute segments, attach aircraft type	AH	One table of segments for January 2026 on the corridor
Interpolate ERA5 to every segment; compute RHi; apply the formation and persistence tests	SV	Every segment has two flags
Run CoCiP through pycontrails on the flagged segments for lifetime and energy forcing	SV	Energy forcing per segment saved; top 20 segments listed
Make 10.8 – 12.0 μm images for the corridor area for every 15-minute slot in three chosen days	IK	Images saved with timestamps
Write labelling guide v0.1 and run the 20-scene pilot, with two labellers per scene	IK, SV	Agreement (Cohen's κ) computed; guide revised
Overlay: draw wind-advected predicted contrails on satellite images for three days	IK, SV	Three clear figures, used in brief, video and website
Dashboard v0: static web map with flagged segments and one example flight card	AS	Opens from a link on a phone and a laptop

What v0 is allowed to be. Rough. It only needs to show that the pipeline runs from raw data to a map, and that predictions can be compared with what the satellite shows. Say plainly in the video that the correction model comes next.

3 Innovation Stage submission

7 Dec 2026 – 7 Jan 2027
internal deadline 5 Jan

GOAL	DELIVERABLE	CHECKPOINT
Submit a brief, video and website that tell one consistent story and score well on all five judging categories.	Innovation Brief with reference attachment, 3–5 minute video (YouTube or Vimeo link plus MP4 under 800 MB), website, fee paid or aid granted.	Everything submitted on 5 Jan, two days before the official deadline.

Holiday window. Shrugal is away from 24 Dec to 2 Jan, and others may travel too. So the full first draft of everything is due on 18 Dec, Dr. Pant reviews on 22–23 Dec, and 3–5 Jan is only for fixes, not new work.

Innovation Brief: who writes what

SECTION	WORDS	OWNER	SOURCE MATERIAL
Elevator Pitch	150	AS	Lean Canvas, tagline
Team	150	AS	Roles table; one line of relevant experience each
Opportunity	300	SV	Synopsis sections 2–3
Key Metrics (design, technology, impact)	750	SV, IK, AH	Paper sections 4 and 6; pipeline figure
Validation / Progress	450	AH	Coverage audit, v0 prototype, overlay images, pilot κ, advisor replies
Market	300	AS	Market sheet
Competition	300	AS	Competitor sheet
Go-To-Market	150	AS	Lean Canvas channels and early adopters
Business Model	300	AS, SV	Cost table, subscription and credit assumptions
Fundraising	150	AS	Grants, competitions, financial aid
Total	**3,000**		Plus reference attachment (SV) and up to two optional attachments: the synopsis and the research paper

Video and website

TASK	OWNER	DONE WHEN
Video script, about 4 minutes: hook with a real contrail photo (0:00–0:30), problem and the India gap (to 1:30), live v0 demo and overlay images (to 3:00), business model and next steps (to 4:00), team (to 4:30)	AS, SV	Script read aloud in under 4:45
Take our own photos or short clips of contrails over Bangalore on clear days	All	At least five usable shots
Record, edit, add captions; upload as unlisted and export MP4 under 800 MB	AS	Link works in a private browser window

TASK	OWNER	DONE WHEN
Website: home, how it works, progress log, team, contact; same name, tagline and numbers as the brief	AS	Live by 22 Dec
Consistency check: every number appears the same in brief, video, website and canvas	SV	Checklist signed off
Entry fee: if we need financial aid, apply early in December so the answer arrives before the deadline	VP, AS	Fee paid or aid confirmed

4 Satellite validation

8 Jan – 28 Feb 2027
about 7 weeks

GOAL	DELIVERABLE	CHECKPOINT
Answer the first research question: how well does raw ERA5 predict contrails that the satellite actually sees over India?	Evaluation report v1 with the result for hypothesis H1 and scores for baselines B0–B3.	28 Feb: report reviewed by Dr. Pant and, ideally, an outside advisor.

TASK	OWNER	DONE WHEN
Extend the pipeline from one month to the full evaluation year, March 2025 – February 2026	AH	Segments for all twelve months
Label 200 scenes chosen across months and times of day, two labellers each	IK + one other	200 scenes labelled; κ reported
Implement matching: advect segments with ERA5 winds; match detections by distance, angle and age; build the observability mask	SV, IK	Each segment is observed, not observed or unknown
Run baselines B0 (climatology), B1 (raw ERA5), B2 (constant scaling) and B3 (pycontrails latitude correction)	SV	Scores with bootstrap intervals for each
Test H1 exactly as pre-registered and write it up, whatever the answer	SV	Result section drafted
When results arrive in February, read the judges' coaching comments together and turn each into a task	All	Comment list in the task board within one week

Minimum sample. If we have fewer than 300 matched contrails in observable scenes, we cannot test H1 or H2 properly. In that case we label more scenes before running the final numbers, and we say so.

5 Correction model and Summit preparation

Feb – 24 Apr 2027
Summit 21–24 Apr

GOAL	DELIVERABLE	CHECKPOINT
Show that observations can correct the forecast (H2), turn it into ranked decisions, and pitch it well in Houston.	Correction model, dashboard v1, Power Pitch, expo booth.	Full pitch rehearsal with Dr. Pant two weeks before travel.

5A. Research and product (all teams, finalists or not)

TASK	OWNER	DONE WHEN
Train the correction model on three seasons, test on the fourth, rotating; compare with B3	SV	H2 result with intervals
Concentration and night analyses for H3 and H4	SV, AH	Two figures and two results
Avoidance options at ±2,000 and ±4,000 ft with OpenAP fuel; rank by lower bound of net benefit	AH	Ranked list for every day in the test season
Dashboard v1: map by flight level, ranked list, flight cards, methods page	AS	Usable by someone who has never seen it

5B. If we are finalists

TASK	OWNER	DONE WHEN
Travel: check passports are valid well beyond April, and start US visa appointments the day finalists are announced, because waiting times in India can be long. Check current waiting times now so there are no surprises.	VP + families	Appointments booked
Power Pitch script and slides; rehearse at least ten times, timed	All	Two clean run-throughs in a row
Hard-question bank: 20 questions judges are likely to ask, with short answers. Start with "Isn't India's contrail problem small?", "What if air traffic control says no?", and "How do you know the satellite really saw that contrail?"	SV, AS	Every member can answer any question
Expo booth: printed poster, printed Lean Canvas, laptop demo running offline, a one-page handout	AS, IK	Packed and tested

5C. If we are not finalists

Read the coaching comments carefully, finish 5A anyway, and point the work at the next opportunities: a preprint, the open dataset, and science fairs in the 2027–28 cycle, such as the IRIS National Fair. A finished, honest result is worth more than a Conrad placing.

6 Publish and continue

May 2027 onward
open-ended

GOAL	DELIVERABLE	CHECKPOINT
Make the work public and useful beyond the competition.	Preprint, open dataset with a permanent identifier, updated dashboard.	Preprint posted and dataset released.

TASK	OWNER	DONE WHEN
Update the research paper with results and post it as a preprint (EarthArXiv, or arXiv if an endorser can be found)	SV, VP	Public link
Release labels, predictions, code and a list of what did not work, with a permanent dataset identifier	AH, IK	Dataset page live
Share results with the researchers and climate groups who replied to our outreach; ask what would make the data useful to them	AS	Three conversations held
Extend from one corridor to the full Indian domain, with a special look at overnight cargo routes	AH, SV	Domain-wide map
Only if the results support it: approach one cargo operator's sustainability team about a no-cost review of past night flights	AS, VP	Meeting held, or decision recorded not to

A Appendix A. Reading list

One paper a week from Phase 1. The presenter gives a 10-minute summary: the question, the method, the main number, and what it means for STRATUM.

WEEK	PAPER	WHY WE READ IT	PRESENTS
1	Teoh et al. (2024), Global aviation contrail climate effects from 2019 to 2021, *ACP*	The only estimate covering India; the humidity correction we test against	SV
2	Sonabend-W et al. (2024), Feasibility test of per-flight contrail avoidance, *Communications Engineering*	Proof that avoidance works in a real airline	AS
3	Schumann (1996), On conditions for contrail formation from aircraft exhausts, *Meteorologische Zeitschrift*	The Schmidt–Appleman criterion properly	SV
4	Driver et al. (2025), Factors limiting contrail detection in satellite imagery, *AMT*	What the satellite can and cannot see	IK
5	Wolf et al. (2025), Correction of ERA5 temperature and humidity biases, *ACP*	How large the ERA5 humidity bias is	AH
6	Santos Gabriel et al. (2026), Adaptive contrail detection in Meteosat imagery, *AMT*	A detector built for our satellite type	IK
7	Teoh et al. (2020), Mitigating contrail forcing by small-scale diversions, *ES&T*	The diversion idea, modelled	AH
8	Stuber et al. (2006), Diurnal and annual cycle of air traffic and contrail forcing, *Nature*	Why night flights matter	AS

B Appendix B. If we fall behind

Cut in this order, and record each cut in the decision log. Never cut the pre-registered hypotheses or the honest write-up.

1. Optional baseline B4 (quantile mapping).
2. The west–east spatial test, keeping the seasonal folds.
3. The ±4,000 ft alternatives, keeping ±2,000 ft.
4. Expanding from the corridor to the full Indian domain.
5. Dashboard polish. A plain page that works beats a beautiful one that does not.

C Appendix C. Risk register

RISK	LIKELIHOOD	WHAT WE DO
OpenSky coverage over India too low	Medium	Decision rule; fall back to ADS-B Exchange days
Contrails hard to see in Meteosat images	Medium	Focus on clear winter scenes; observability mask; more labels
No advisor replies	Medium	Follow up after one week; widen the list; ask Dr. Pant for introductions
Holiday clash with the 7 Jan deadline	High	Draft complete by 18 Dec; final fixes only from 3 Jan
Results do not support H1 or H2	Possible	Publish anyway; shift the story to H3 and H4
Fee or travel cost	Medium	Apply for financial aid early; look for school or local sponsorship
Visa delays if finalists	Medium	Check waiting times now; book the day results come out

Use of AI tools. We used an AI assistant to help organise this plan and to check the competition requirements against the official Student Guide. Dates and requirements must be rechecked against the guide before each deadline.